

Abstract No.

Myelin-driven lipid metabolism reprograms macrophages to suppress anti-tumor immunity in brain metastasis

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Background: Modulation of lipid metabolism through myelin recycling has recently emerged as a critical driver of macrophage-mediated immunosuppression and tumor progression in glioblastoma. However, the underlying mechanisms in brain metastases remain poorly characterized.

Methods: We aimed to reprogram the immunosuppressive tumor microenvironment in experimental brain metastasis models by combining radiotherapy with either adenosine signalling blockade or immune checkpoint inhibition. Comprehensive immunophenotyping including multiplexed histology, flow cytometry, single-cell RNA sequencing, spatial transcriptomics, and metabolomics was performed to delineate immunomodulatory effects. Macrophages in ex vivo Cancer Core Models, consisting brain metastatic tissue embedded in cerebral organoids, were stimulated with myelin to induce lipid-laden macrophage (LLM) and assess functional consequences for cancer cell proliferation, macrophage activation, and T cell responses using imaging and transcriptomic profiling. Macrophages were further modulated by supplementing adenosine or inosine, as well as by neutralizing antibodies targeting cell–cell communication between LLMs and T cells.

Results: In mouse breast-to-brain metastasis, radiotherapy induced a pronounced expansion of lipid-laden macrophages characterized by immunosuppressive polarization, reduced antigen presentation, and impaired T cell activation. Cell–cell interaction analyses revealed macrophage-mediated FN1 and SPP1 suppressive signaling to limit T cell responses and therapeutic efficacy. In contrast, combining radiotherapy with adenosine signaling blockade prevented LLM accumulation and promoted expansion of antigen-presenting macrophages, resulting in significant survival benefits. Ex vivo models recapitulated the induction of LLMs and confirmed their immunosuppressive function. Notably, macrophage modulation exhibited heterogeneity across different brain metastatic origins, including lung cancer and melanoma. Metabolomic profiling of breast-to-brain metastasis treated with radiotherapy and/or immunotherapy revealed distinct alterations in metabolites and lipid species critical for macrophage function, including adenosine, sphingomyelin, ceramide, and phosphatidylcholine.

Conclusion: Targeted modulation of lipid metabolism in tumor-associated macrophages represents a promising therapeutic strategy to alleviate immunosuppression and enhance anti-tumor immunity in brain metastases.



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